Statistical Inference and Modeling

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Lectures

12. Support Vector Machine

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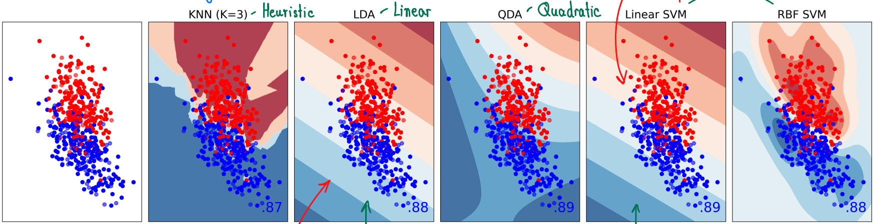
Support Vector Machine (SVM) vs. Others

↳ สามารถปรับ config (Kernal) ได้โคตรเยอะ + จินตนาการการตัดได้

Relax Assumption. base on Kernal.



Normal Assumption

KNN

- Identify the class that is the majority of the k closest training features.

Logistic Regression

- Mapping an input with a sigmoid function with a set of parameters β.
- β is from solving a maximum likelihood of $p(y_c|x_c, \beta)$.

LDA, QDA

- Mapping an input with a linear/quadratic function with a set of parameters β.
- β is from the mean and covariance of training statistics for $p(x_c|y_c)$

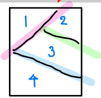
สังเกตว่ามันให้ผลคล้ายกัน
เพราะ obj f^n: $f(x^j) = w^T x^j + b$
(เป็น linear) ถ้า $X \overset{iid}{\sim} \mathcal{N}(\mu_c, \Sigma)$
#ข้อมูลนี้มัน generate โดย Normal ทำให้ได้ผลคล้ายกัน.

SVM

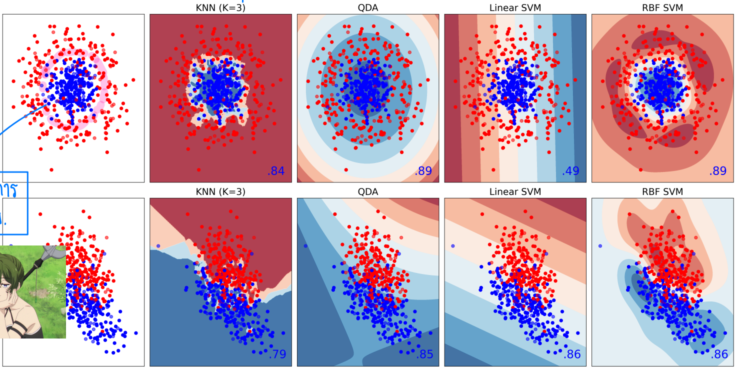
- Mapping an input with a function with a set of parameters β.
- β from finding a hyperplane that best separate the training data.
- Hyperplane will capture the interaction between $\{x_c\}$ vs. $\{x_k\}_{k \neq c}$

Why SVM?

Class 1 และไม่ใช่ 1



ถ้าเป็น Multi-class แล้ว
SVM จะทำการไปตัดแบบ 1 vs other ไปเรื่อยๆ



ถ้าเราจินตนาการ
ว่าตัดได้ = จบ.

- **When SVM could be the better choice?**
  - When you found that the interaction between classes can be useful.
- **How SVM models such interaction?**
  - Through the construction of the hyperplanes.
  - The parameters in hyperplane are learned based on the interaction between classes.
  - Such hyperplanes are used to perform classification.

So, what is SVM ? (Binary Classification)

- **Training**: solve the following optimization problem ... data sample.

$$\boxed{\hat{b}, \hat{\boldsymbol{w}} = \underset{b \in \mathcal{R}, \boldsymbol{w} \in \mathcal{R}^n}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 \quad \text{s.t.} \quad y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1 \text{ for } j = 1, 2, ..., m}$$

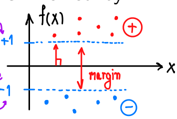
- **Inference**: given a n-dimensional input feature $\boldsymbol{x}_j \in \mathcal{R}^n$, the class label is inferred by

Normal Vector ของ Hyperplane

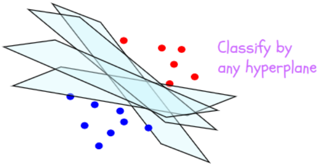
$$f(\boldsymbol{x}_j) = \hat{\boldsymbol{w}}^T\boldsymbol{x}_j + b$$

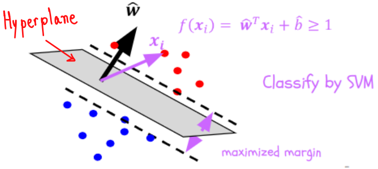
data sample

bias ของ hyperplane

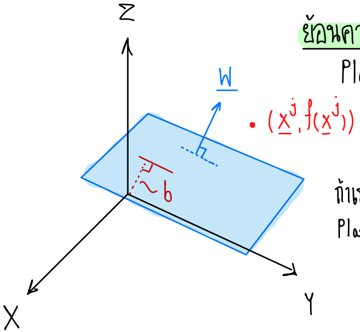


where $\hat{b} \in \mathcal{R}, \hat{\boldsymbol{w}} \in \mathcal{R}^P$ are learned from the training phase.





Idea หยาบๆ → แนวคิดพื้นฐานของ SVM



ย้อนความทรงจำ ตอน Cal 2 :

Plane eq$^{\underline{n}}$: $w_1x_1 + w_2x_2 + \ldots + w_nx_n = d$

$f(\underline{x}) = 0$: $\underline{w}^T\underline{x} + \underline{b} = 0$, $f : \mathbb{R}^{n\times n} \longrightarrow \mathbb{R}^{n\times 1}$

bias

ถ้าเราบอกว่า $f(\underline{x}^j) = \underline{w}^T\underline{x}^j + b > 0$ แสดงว่า $\{(\underline{x}^j, f(\underline{x}^j))\}$ มันก็ควรจะอยู่บน Plane นั้นๆ หรือมีทิศทางเดียวกับ Normal Vector $\underline{w}$ $\therefore y_j = 1$

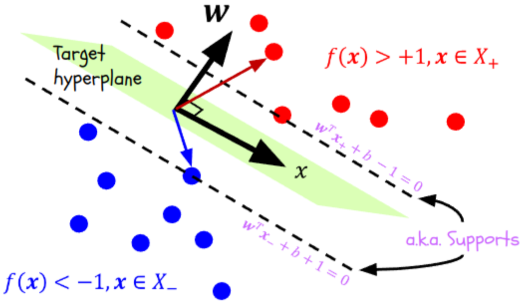
นิยาม dot

→ จุดที่ตรงข้าม $\underline{w}^Tx^j + b < 0$ แปลว่ามันทิศตรงข้าม ⇒ $y_j = -1$

$\vec{u} \cdot \vec{v} = \|\vec{u}\|_2 \cdot \|\vec{v}\|_2 \cdot \cos\theta$

ระบุทิศทาง

Supports are the points lying on the two additional, parallel hyperplanes



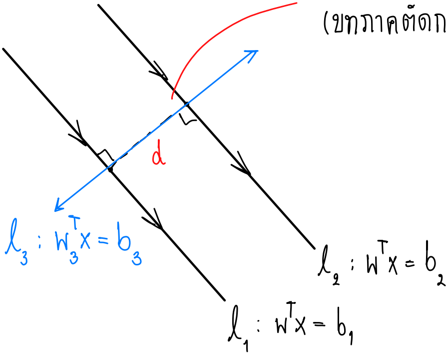
$f(x_+) = w^T x_+ + b = 1$

- $\boxed{\boldsymbol{w}^T \boldsymbol{x}_+ + b - 1 = 0}$
- $\boldsymbol{w}^T \boldsymbol{x}_- + b + 1 = 0$

positive $\underbrace{\text{support hyperplane.}}_{\text{boundary.}}$

สมบัติ ① . Target hyperplane ต้อง parallel กับ support hyperplane

② . " ______________ " ทำให้ระยะ 2 support ให้ได้มากสุด.



ตอนเรียน ม.4 (บทภาคตัดกรวย) : $d = \dfrac{|b_1 - b_2|}{\sqrt{A^2 + B^2}}$

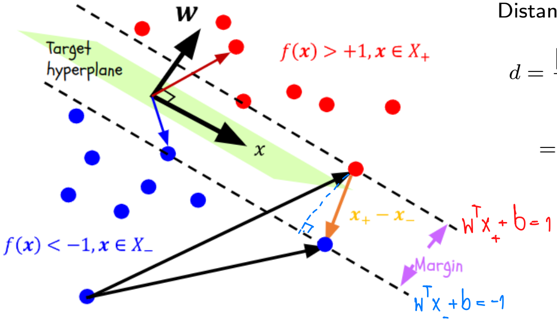
$$\boxed{d = \frac{|b_1 - b_2|}{\|w\|_2}}$$

* เส้น $l_1 \parallel l_2$ แต่ $\perp$ l_3

$\therefore w_{l_3} \cdot w = 0$ (dot product).

Find the supports that maximize the margin

Our goal is to find $b \in \mathcal{R}, \boldsymbol{w} \in \mathcal{R}^n$ that maximizes the margin between two supports.



Distance between supports

$$d = \frac{|\boldsymbol{w}^T(\boldsymbol{x}_+ - \boldsymbol{x}_-)|}{||\boldsymbol{w}||_2}$$

$$= \frac{|1 - b - (-1 - b)|}{||\boldsymbol{w}||_2} = \frac{2}{||\boldsymbol{w}||_2}$$

"w" will decide how wide the margin will be ...

Maximizing $\frac{2}{||\boldsymbol{w}||_2} \triangleq$ Minimizing $||\boldsymbol{w}||_2$. → obj f^n

Constraints of how to map the classes in training data

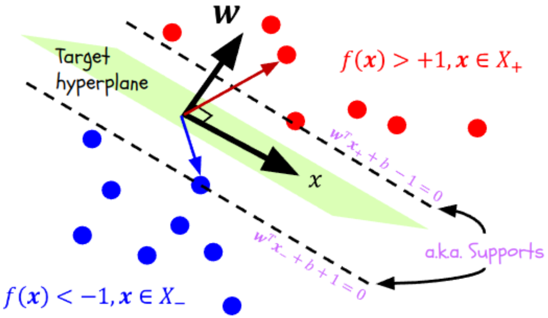
Given m pairs of training data $\{\boldsymbol{x}_j, y_j\}$ where

- $\boldsymbol{x}_j \in \mathcal{R}^n$ is the input feature;
- y_j is the associated label $\in \{-1, 1\}$.

if-else constraint :

For $\boldsymbol{x}_j \in X_+ \quad \Rightarrow \quad y_j = +1 \;\&\; f(\boldsymbol{x}_j) > +1 \quad \Rightarrow \quad y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1$

For $x_j \in X_- \quad \Rightarrow \quad y_j = -1 \;\&\; f(\boldsymbol{x}_j) < -1 \quad \Rightarrow \quad y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1$



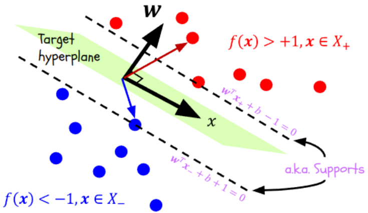
if $\exists\{x_j, y_j\}$ can not be linearly seperated
then no solution for $\hat{w}, \hat{b}$.

SVM learning is performed by solving the optimization problem ...

$$\hat{b}, \hat{\boldsymbol{w}} = \underset{b\in\mathcal{R}, \boldsymbol{w}\in\mathcal{R}^n}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 \quad \text{s.t.} \quad y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1 \text{ for } j = 1, 2, ..., m.$$

The problem is a convex quadratic program where $b \in \mathcal{R}, \boldsymbol{w} \in \mathcal{R}^n$ are chosen such that

- they agree with the class mapping constraints $y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1$; and
- they are minimizing $||\boldsymbol{w}||_2$ (as a way of maximizing $\frac{2}{||\boldsymbol{w}||_2}$).



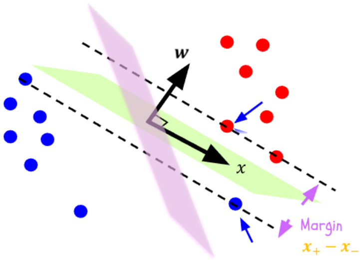
SVM learning — the comments on hard margin SVM

SVM Learning is performed by solving the optimization problem ...

$$\hat{b}, \hat{\boldsymbol{w}} = \underset{b\in\mathcal{R}, \boldsymbol{w}\in\mathcal{R}^n}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 \quad \text{s.t.} \quad y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1 \text{ for } j = 1, 2, ..., m.$$

Although the constraint helps putting $\{\boldsymbol{x}_j, y_j\}$ in the correct class mapping, it also enforces that $b, \boldsymbol{w}$ are the one **that linearly separates the data.**

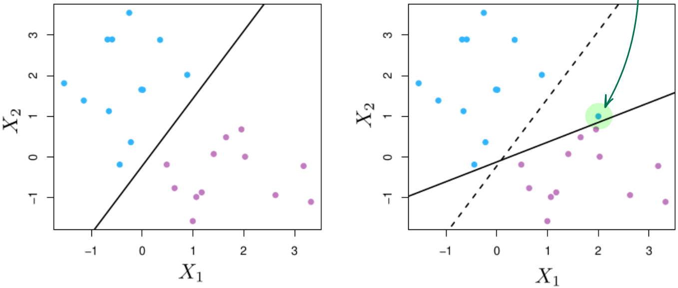
- This version of SVM is sometimes called **hard margin**.
- If **the data cannot be linearly separated**, which is possible due to noise, **the accuracy of the classification can be compensated/reduced**.



You may miss out a better way to separate the class.

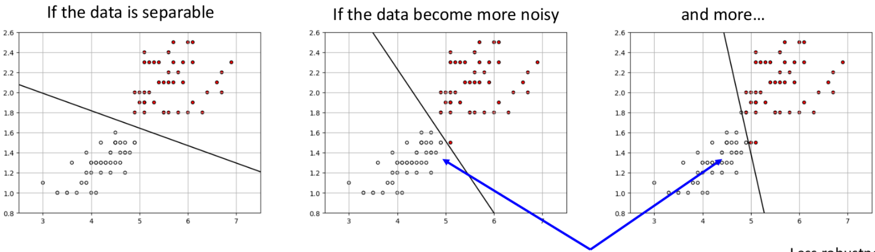
What is the hard margin's problem?

จุดที่ทำให้ hyperplane มัน shift มากๆ (Fit to noise).



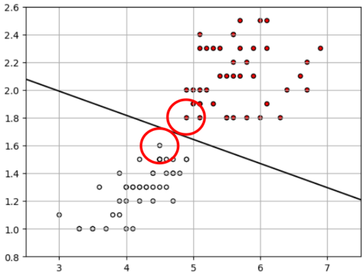
- *From left to right.* By adding an **odd** (noise) pair of data, the hyperplane dramatically changes; it may overfit to the noise in training data.
- Hard (linear) decision on the margin is no longer useful — we need something more robust against such noise.
- A soft decision to ignore the **noise**, so the hyperplane will not try to fit to **the odd pair**, but it aims at grouping data in overall.

Impact of noise on hard margin

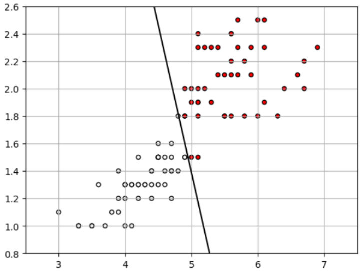


The hyperplane will be fitted to the noisy data…… ➡ ***The hyperplanes*** are **tilted** because they are compensating with the noisy (non-separable) data. ➡ Less robustness against noise when inference…

Clean training features



Noisy training features

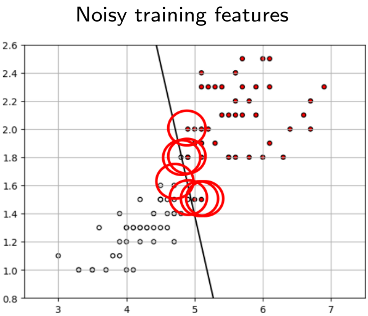


Red ball represents the smallest shifting that can be misclassified by the hyperplane on **the left**.

Impact of noise on hard margin

- Apply the same red ball to noisy data on **the right**...
- **On the right**, many red balls are overlapped with the hyperplane.
  It is **more likely** to misclassify, even if some of the data can be correctly classified by the hyperplane on the left.

Summary



- Noise in our training samples $\Rightarrow$ an extremely tilt hyperplane (on the right)
- Extremely tilt hyperplane $\Rightarrow$ very likely to misclassify $\Rightarrow$ **low robustness**.

The hard margin is said to have **low robustness** against noisy training samples.

force ให้ Slack มันอย่ามากไป

$$\hat{b}, \hat{\boldsymbol{w}} = \underset{b \in \mathcal{R}, \boldsymbol{w} \in \mathcal{R}^n, \boldsymbol{z} \in \mathcal{R}^m}{\textbf{argmin}} \ \frac{1}{2}||\boldsymbol{w}||_2^2 + \boxed{C \sum_{j=1}^{m} z_j}$$

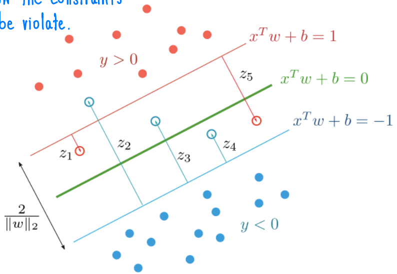
$$\text{s.t.} \quad y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) \geq 1 - z_j,$$

$$z_j \geq 0$$

Allow the constraints to be violate.

Slack Variable

$$\text{for } j = 1, 2, ..., m$$



- z_j is called a slack variable.
- If $z_j > 0$ at optimum, the mapping of $\boldsymbol{x}_j$ is relaxed from the primal condition, which allows the inclusion of $\boldsymbol{x}_j$ that is on the wrong side of the hyperplane.
- The regularization parameter C controls the trade-off between **maximizing the margin** and **the distance to the opposite side**.

Soft margin: varying parameter C

Margin = $\frac{2}{||w||_2}$

$$\hat{b}, \hat{\boldsymbol{w}} = \underset{b\in\mathcal{R}, \boldsymbol{w}\in\mathcal{R}^n, \boldsymbol{z}\in\mathcal{R}^m}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 + C\sum_{j=1}^{m} z_j$$

s.t. $y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1 - z_j$, and $z_j \geq 0$ for $j = 1, 2, ..., m$

Small C

$\sum_j z_j$ is less penalized

z_j is large

Relaxing $y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1$

the margin $(1/||\boldsymbol{w}||_2)$ is large

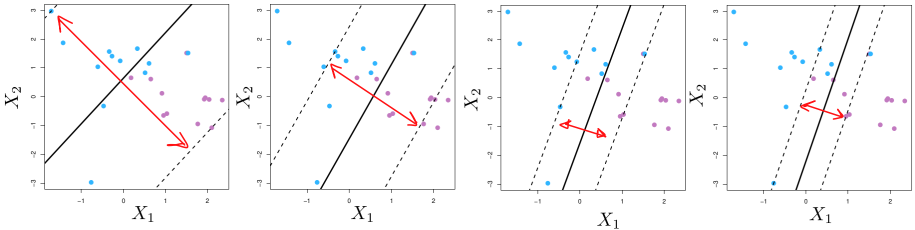
Large C

$\sum_j z_j$ is more penalized

z_j is small

Enforcing $y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1$

the margin $(1/||\boldsymbol{w}||_2)$ is small



Keywords

- Margin
- Supports
- Hard margin
- Soft margin
- Slack variables z_i
- Regularization parameter C
- Is there anything remind you of LDA ...? $f(x_j) = w^T x_j + b$, if X_j is normally distributed.

Dual problem

Primal Problem

Training data: data pairs $\{\boldsymbol{x}_j, y_j\}$

Convex f^n

$$\underset{b\in\mathcal{R}, w\in\mathcal{R}^n, z\in\mathcal{R}^m}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 + C\sum_{j=1}^{m} z_j$$

Decision Variables.

$$\text{s.t.} \quad y_j(\boldsymbol{x}_j^T\boldsymbol{w} + b) \geq 1 - z_j,$$

$$z_j \geq 0$$

for $j = 1, ..., m$

Inference: given an input $\boldsymbol{x}$

$$f(\boldsymbol{x}) = \boldsymbol{x}^T\hat{\boldsymbol{w}} + \hat{b}$$

- $\hat{\boldsymbol{w}}, \hat{b}$ are learned parameters.

note : $y_j \in \{-1, 1\}$; $\forall j$ ⇒ เราต้องรู้ y_j ก่อน

* SVM ไม่ใช่ Clustering Algorithm.

Dual Problem

Training data: data pairs $\{\boldsymbol{x}_j, y_j\}$

C เรากำหนดเอง (hyper parameter) ⇒ การ Allowance ข้ามฝั่ง.

$$\underset{\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathcal{R}^n}{\textbf{argmax}} \; 1^T\boldsymbol{\alpha} - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j x_i^T x_j$$

$$\text{s.t.} \quad \sum_{j=1}^{m}\alpha_j y_j = 0,$$

$$\alpha_j = C - \beta_j, \quad \alpha_j, \beta_j \geq 0$$

for $j = 1, ..., m$

Inference: given an input $\boldsymbol{x}$

$$f(\boldsymbol{x}) = \boldsymbol{x}^T\hat{\boldsymbol{w}} + \hat{b}$$

- $\hat{\boldsymbol{w}} = \sum_{j=1}^{m}\hat{\alpha}_j y_j \boldsymbol{x}_j$ where each pair $\{\boldsymbol{x}_j, y_j\}$ are the training data. $\hat{\alpha}_i$ often takes 0 value.
- $\hat{b}_j = y_j - \hat{\boldsymbol{w}}^T\boldsymbol{x}_j$ for each pair $\{\boldsymbol{x}_j, y_j\}$.

What is the (dual) problem formulation?

Primal Problem

Training data: data pairs $\{\boldsymbol{x}_j, y_j\}$

$$\underset{b\in\mathcal{R},\boldsymbol{w}\in\mathcal{R}^n,\boldsymbol{z}\in\mathcal{R}^m}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 + C\sum_{j=1}^{m} z_j$$

$$\text{s.t.} \quad y_j(\boldsymbol{w}^T\boldsymbol{x}_j + b) \geq 1 - z_j,$$

$$z_j \geq 0$$

$$\text{for } j = 1, ..., m$$

Inference: given an input $\boldsymbol{x}$

$$f(\boldsymbol{x}) = \boldsymbol{x}^T\hat{\boldsymbol{w}} + \hat{b}$$

Dual Problem

Training data: data pairs $\{\boldsymbol{x}_j, y_j\}$

$$\underset{\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathcal{R}^m}{\textbf{argmax}} \ 1^T\boldsymbol{\alpha} - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \boldsymbol{x}_i^T\boldsymbol{x}_j$$

ส่วนใหญ่ take zero

$$\text{s.t.} \quad \sum_{j=1}^{m}\alpha_j y_j = 0,$$

$$\alpha_j = C - \beta_j, \quad \alpha_j, \beta_j \geq 0$$

$$\text{for } j = 1, ..., m$$

Inference: given an input $\boldsymbol{x}$

$$f(\boldsymbol{x}) = \boldsymbol{x}^T\hat{\boldsymbol{w}} + \hat{b}$$

- $\hat{\boldsymbol{w}} = \sum_{j=1}^{m}\hat{\alpha}_j y_j \boldsymbol{x}_j$ where each pair $\{\boldsymbol{x}_j, y_j\}$ are the training data. $\hat{\alpha}_j$ often takes 0 value.
- $\hat{b}_j = y_j - \boldsymbol{x}_j^T\hat{\boldsymbol{w}}$ for each pair $\{\boldsymbol{x}_j, y_j\}$.

Why the (dual) problem formulation?

Primal Problem

Inference: given an input $\boldsymbol{x}$

$$f(\boldsymbol{x}) = \boldsymbol{x}^T \hat{\boldsymbol{w}} + \hat{b}$$

Dual Problem

Inference: given an input $\boldsymbol{x}$

$$f(\boldsymbol{x}) = \boldsymbol{x}^T \hat{\boldsymbol{w}} + \hat{b}$$

- $\hat{\boldsymbol{w}} = \sum_{j=1}^{m} \hat{\alpha}_j y_j \boldsymbol{x}_j$ where each pair $\{\boldsymbol{x}_j, y_j\}$ are the training data.
- $\hat{\alpha}_j$ often takes 0 value. (แปลว่าเรา map ถูกด้าน)
- Once $\hat{\boldsymbol{\alpha}}_j$ is given, you can select which $\{\boldsymbol{x}_j, y_j\}$ are going to be used for estimating $\hat{b}$ and $\hat{\boldsymbol{w}}$.

... First reason is efficiency.

Second reason is higher order.

Linear mapping

การ dot ของ feature vector.

Training data: data pairs $\{\boldsymbol{x}_j, y_j\}$

$$\underset{\alpha,\beta\in\mathcal{R}^n}{\textbf{argmax}}\ 1^T\alpha - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \boldsymbol{x}_i^T \boldsymbol{x}_j$$

$$\text{s.t.}\quad \sum_{j=1}^{m}\alpha_j y_j = 0,$$

$$\alpha_j = C - \beta_j,\quad \alpha_j, \beta_j \geq 0$$

for $j = 1, \ldots, m$

Inference: given an input x

$$f(\boldsymbol{x}) = \underbrace{\sum_{j=1}^{m}\hat{\alpha}_j y'_j \boldsymbol{x}'_j \boldsymbol{x}}_{\boldsymbol{w}^T\boldsymbol{x}} + \hat{b}$$

- the pair $\{y'_j, \boldsymbol{x}'_j\}$ are of the training data.
- $y_j(\hat{b} + \boldsymbol{x}_j^T\hat{\boldsymbol{w}}) = 1$ for each pair $\{\boldsymbol{x}_j, y_j\}$.

Kernel mapping

Kernel function.

Training data: data pairs $\{\boldsymbol{x}_j, y_j\}$

$$\underset{\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathcal{R}^m}{\textbf{argmax}}\ 1^T\alpha - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

$$\text{s.t.}\quad \sum_{j=1}^{m}\alpha_j y_j = 0,$$

$$\alpha_j = C - \beta_j,\quad \alpha_j, \beta_j \geq 0$$

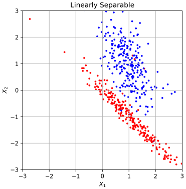
for $j = 1, \ldots, m$

Inference: given an input x

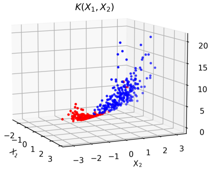
$(x_j^T \underline{x})$

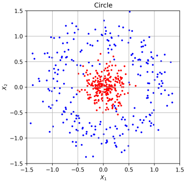
$$f(\boldsymbol{x}) = \underbrace{\sum_{j=1}^{m}\hat{\alpha}_j y'_j \mathcal{K}(x'_j, \boldsymbol{x})}_{\boldsymbol{w}^T\boldsymbol{x}} + \hat{b}$$

- the pair $\{y'_j, \boldsymbol{x}'_j\}$ are of the training data.
- $\hat{b} = \frac{1}{m}\sum_j(y_j - \boldsymbol{x}_j^T\hat{\boldsymbol{w}})$ for each $\{\boldsymbol{x}_j, y_j\}$.

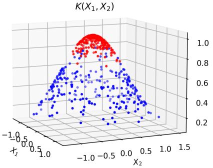


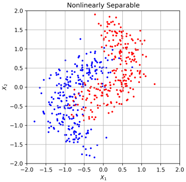
Polynomial $\Rightarrow$ $$\mathcal{K}(\boldsymbol{x}_1, \boldsymbol{x}_2) = (\gamma \boldsymbol{x}_1^T \boldsymbol{x}_2 + \beta)^d$$ $\Rightarrow$



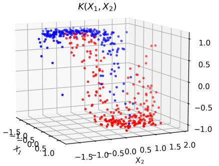


Radial Basis $\Rightarrow$ $$\mathcal{K}(\boldsymbol{x}_1, \boldsymbol{x}_2) = e^{-\gamma ||\boldsymbol{p} - \boldsymbol{x}||_2^2}$$ $\Rightarrow$





Tanh $\Rightarrow$ $$\mathcal{K}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \tanh(\gamma \boldsymbol{x}_1^T \boldsymbol{x}_2 + \beta)$$ $\Rightarrow$



because $\alpha > 0$

$f(x) \geq f(x) - \alpha \cdot g(x) \rightarrow f(x) = \max(f(x) \cdot g(x))$

Came from the primal's obj

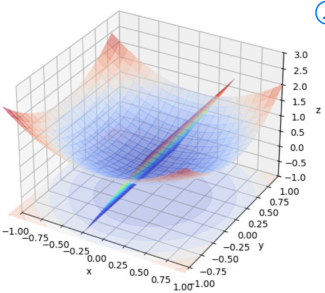
Primal problem $\Rightarrow$ Lagrangian $\Rightarrow$ Dual problem

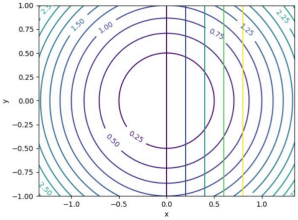
$$\underset{\boldsymbol{x} \in \mathbb{R}^N}{\text{minimize}}\, f(\boldsymbol{x}) \quad \text{subject to} \quad g(\boldsymbol{x}) \geq 0$$

$$L(\boldsymbol{x}, \boldsymbol{\alpha}) = f(\boldsymbol{x}) - \alpha\, g(\boldsymbol{x}) \quad \text{subject to} \quad \alpha \geq 0$$

$$\underset{\boldsymbol{\alpha} \in \mathbb{R}}{\text{maximize}}\ \inf_{\boldsymbol{x} \in \mathbb{R}^N} L(\boldsymbol{x}, \boldsymbol{\alpha}) \quad \text{subject to} \quad \alpha \geq 0$$

1. $f(x) \geq f(x) - \alpha g(x)$; $\alpha \geq 0$, $g(x) \geq 0$
2. $\nabla f(x^*) = \alpha \cdot \nabla g(x^*)$





$$\mathcal{L}(\underline{x}, \alpha) = f(\underline{x}) - \alpha \cdot g(\underline{x})$$

Primal problem $\Rightarrow$ **Lagrangian**

$$\underset{b\in\mathcal{R},\boldsymbol{w}\in\mathcal{R}^n,\boldsymbol{z}\in\mathcal{R}^m}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 + C\sum_{j=1}^{m} z_j$$

$$\left.\begin{aligned} \text{s.t.} \quad & y_j(\boldsymbol{x}_j^T\boldsymbol{w} + b) - 1 + z_j \geq 0, \\ & z_j \geq 0 \end{aligned}\right\} \text{มี } g(x) \text{ 2 ชุด}$$

for $j = 1, ..., m$

$$L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{2}||\boldsymbol{w}||_2^2 + C\sum_{j=1}^{m} z_j - \sum_{j=1}^{m} \alpha_j(y_j(\boldsymbol{x}_j^T\boldsymbol{w} + b) - 1 + z_j) - \sum_{j=1}^{m} \beta_j z_j$$

s.t. $\alpha_j, \beta_j \geq 0$ for $j = 1, ..., m$

- α, β are Lagrangian multipliers.

$$\boxed{\begin{aligned} \therefore \text{Dual} : \ & \underset{\alpha,\beta\in\mathbb{R}}{\text{maximize}} \ \underset{b,w,z\in\mathbb{R}}{\inf} \mathcal{L}(\cdot\cdot) \\ & \text{s.t.} \quad \alpha \geq 0, \ \beta \geq 0 \end{aligned}}$$

Derivation from Lagrangian to dual problem

$$L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{2}||\boldsymbol{w}||_2^2 + C\sum_{j=1}^{m} z_j - \sum_{j=1}^{m} \alpha_j (y_j(x_j^T w + b) - 1 + z_j) - \sum_{j=1}^{m} \beta_j z_j$$

$L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ is linear in $\boldsymbol{\alpha}, \boldsymbol{\beta}$. But, $L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ is quadratic in $\boldsymbol{w}$.

$b \neq 0$

$\frac{\partial L}{\partial b}$ 1). $\inf_b L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) \Rightarrow \sum_{j=1}^{m} \alpha_j y_j b \Rightarrow \boxed{\sum_{j=1}^{m} \alpha_j y_j = 0}$

$\frac{\partial L}{\partial z}$ 2). $\inf_{\boldsymbol{z}} L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) \Rightarrow \sum_{j=1}^{m} C z_j - \sum_{j=1}^{m} \alpha_j z_j - \sum_{j=1}^{m} \beta_j z_j \Rightarrow \boxed{(C - \alpha_j - \beta_j) = 0}$

$\frac{\partial L}{\partial W}$ 3). $\inf_{\boldsymbol{w}} L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) \Rightarrow \partial_{w_j} L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \boldsymbol{0}$

$$\boxed{w_i = \sum_{j=1}^{m} \alpha_j y_j x_{ji}} \Rightarrow \boldsymbol{w} = [w_i]_i$$

$$\inf_{\boldsymbol{w}} L(b, \boldsymbol{w}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{2}||\boldsymbol{w}||_2^2 - \boldsymbol{w}^T d = -\frac{1}{2}||\boldsymbol{w}||_2^2 \text{ where } d = \left[\sum_{j=1}^{m} \alpha_j y_j x_{ji}\right]_i$$

Dual problem

Dual form

$$\underset{\alpha,\beta\in\mathcal{R}^m}{\textbf{argmax}} \inf_{b,\boldsymbol{w},\boldsymbol{z}} L(b,\boldsymbol{w},\boldsymbol{z},\boldsymbol{\alpha},\boldsymbol{\beta})$$

s.t. $\alpha_j, \beta_j \geq 0$ for $j = 1, ..., m$

Dual problem

$$\underset{\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathcal{R}^m}{\textbf{argmax}} \ 1^T\boldsymbol{\alpha} - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \boldsymbol{x}_i^T\boldsymbol{x}_j$$

$$\text{s.t.} \quad \sum_{j=1}^{m}\alpha_j y_j = 0,$$

$$C - \alpha_j - \beta_j = 0,$$

$$\alpha_j, \beta_j \geq 0$$

for $j = 1, ..., m$

Note. Inference step: we still need to compute. $f(\boldsymbol{x}_{\text{input}}) = \hat{\boldsymbol{w}}^T\boldsymbol{x}_{\text{input}} + \hat{b}$

- $\hat{\boldsymbol{w}} = \sum_{j=1}^{m}\hat{\alpha}_j y_j \boldsymbol{x}_j$
- $\hat{b} = \frac{1}{m}\sum_j(y_j - \boldsymbol{x}_j^T\hat{\boldsymbol{w}})$ for each pair $\{\boldsymbol{x}_j, y_j\}$.

Karush–Kuhn–Tucker (KKT) conditions in SVM

KKT conditions are (first-order) **necessary conditions** for **a solution**, in **nonlinear** programming to be optimal, that is,

- If $\hat{p} := \{\hat{\boldsymbol{w}}, \hat{b}, \hat{\boldsymbol{z}}\}$ is a local optimum; then there exist $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ such that the following 4 groups of conditions hold:

  Primal feasibility

$$h_j(\hat{p}) = y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) - 1 + z_j \geq 0$$

$$g_j(z_i) = z_j \geq 0$$

  Stationarity (zero-gradient): $\nabla f(x^*) = \alpha \cdot \nabla g(x^*)$

$$\partial f(\hat{p}) - \sum_j \alpha_j \partial h_j(\hat{p}) - \sum_j \beta_j \partial g_j(z_j) = 0$$

  Complementary slackness สมบัติสำคัญของ Duality → การที่ obj fⁿ ของ Primal = Dual

$$\alpha_j h_j(\hat{p}) = \alpha_j(y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) - 1 + z_j) = 0$$

$$\beta_j g_j(z_j) = \beta_j z_j = 0$$

  Dual feasibility

$$\sum_{j=1}^{m} \alpha_j y_j = 0$$

$$C - \alpha_j - \beta_j = 0 \quad \& \quad \alpha_j, \beta_j \geq 0 \Rightarrow 0 \leq \alpha_j \leq C.$$

Dual feasibility and complementary slackness characterize **three groups of points**.

Beginning	$\longrightarrow$	**End**
Dual feasibility	Complimentary slakness	Primal feasibility
$0 \leq \alpha_j \leq C, \ \alpha_j = C - \beta_j$	$\beta_j z_j = 0, \alpha_j h_j(\hat{p}) = 0$	$y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) \geq 1 - z_j$

1. Correct side of the margin $(z_j \geq 0)$

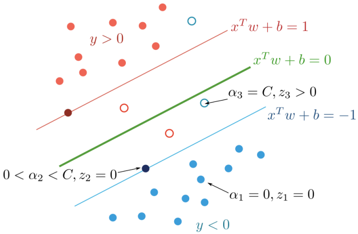
$\alpha_j = 0 \Rightarrow \beta_j = C$	$\beta_j z_j = 0, \ \therefore z_j = 0$	$y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) \geq 1$

2. Edge side of the margin

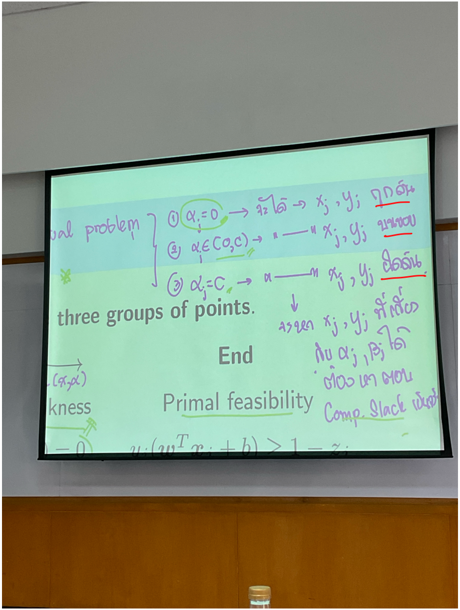
$0 < \alpha_j < C \Rightarrow \alpha_j, \beta_j > 0$	$z_j = 0$	$y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) = 1$

3. Wrong side of the margin

$\alpha_j = C \Rightarrow \beta_j = 0$	$z_j > 0$	$y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) = 1 - z_j$



Three groups of points



$\boxed{\text{Complimentary Slackness.}}$

$\alpha_j = C - \beta_j$; $0 \le \alpha_j \le C$ $\wedge$ $\beta_j z_j = 0$

$y_j(W^T \underline{x}_j + b) \ge 1 - z_j$; $y_j \in \{-1,1\}$

เราใช้ index j ⇒ เรามองแต่ละ data point.

* แปลว่าถ้า $\alpha_j \in [0,C)$, $\alpha_j \ne C$

→ $\{\underline{x}_j, y_j\}$ การันตีการไปอยู่ถูกฝั่งแน่ๆ

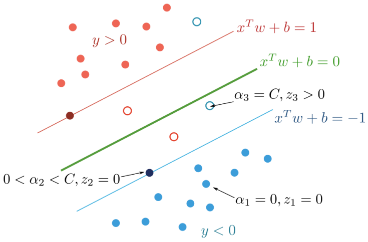
โดย model ที่ดี $\boxed{\alpha_j \text{ ควรเป็น } 0 \text{ เยอะๆ}}$

↓

เพราะมันอยู่ถูกด้าน

ปล. SVM ยังใช้อยู่กับ data sample น้อยๆ

Beginning	$\longrightarrow$	**End**
Dual feasibility	Complimentary slakness	Primal feasibility
$0 \leq \alpha_j \leq C, \ \alpha_j = C - \beta_j$	$\beta_j z_j = 0, \alpha_j h_j(\hat{p}) = 0$	$y_j(w^T x_j + b) \geq 1 - z_j$
1. Correct side of the margin		
$\alpha_j = 0 \Rightarrow \beta_j = C$	$\beta_j z_j = 0, \ \therefore z_j = 0$	$y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) \geq 1$
2. Edge side of the margin		
$0 < \alpha_j < C \Rightarrow \alpha_j, \beta_j > 0$	$z_j = 0$	$y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) = 1$
3. Wrong side of the margin		
$\alpha_j = C \Rightarrow \beta_j = 0$	$z_j > 0$	$y_j(\boldsymbol{w}^T \boldsymbol{x}_j + b) = 1 - z_j$

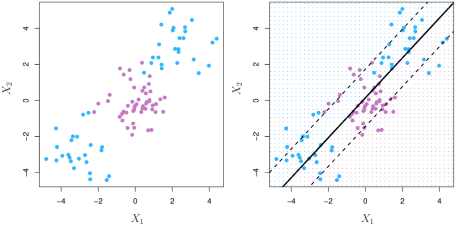


The data sample $\boldsymbol{x}_j$ on the edges (supports) and the wrong sides are the one that contributes to the supports, *i.e.*, $\hat{\boldsymbol{w}} = \sum_{j=1}^{m} \hat{\alpha}_j y_j \boldsymbol{x}_j$, but not many points are around these area.

ถ้า $\alpha_j = 0$ เยอะๆ = Good!

Kernel

Sometimes we face with nonlinear class boundaries and SVM may perform poorly:



Instead of fitting SVM directly, we could map input using :

Augment term

$$\boldsymbol{x} = [x_1, ..., x_n, x_1^2, ..., x_n^2]^T \rightarrow \text{input features.}$$

or, the input vector $\boldsymbol{x} \in \mathcal{R}^{n\times 1}$ can be mapped into an enlarged space $\mathcal{R}^{d\times 1}$ where $d \geq n$:

ใส่โคตร Augment ไป

ตัวอย่างการทำ Augment feature.

$$\boldsymbol{f} = [h(x_1), ..., h(x_n), h(x_1^2), ..., h(x_n^2)]^T$$

Then, the product between $\boldsymbol{f}_j$ and $\boldsymbol{f}_k$ can be represented by a kernel function

$$K(\boldsymbol{x}_j, \boldsymbol{x}_k) = \boldsymbol{f}_j{}^T \boldsymbol{f}_k : \mathcal{R}^{d\times d} \Rightarrow \mathcal{R}^{1\times 1} \quad \text{scalar} \geq 0$$

feature dimension

.

Primal problem

$$\underset{b\in\mathcal{R},\boldsymbol{w}\in\mathcal{R}^n,\boldsymbol{z}\in\mathcal{R}^m}{\textbf{argmin}} \frac{1}{2}||\boldsymbol{w}||_2^2 + C\sum_{j=1}^{m} z_j$$

$$\text{s.t. } y_j({\boldsymbol{f}^j}^T\boldsymbol{w} + b) - 1 + z_j \geq 0,$$

$$z_j \geq 0$$

$$\text{for } j = 1, ..., m$$

Dual (Kernel) problem

$$\underset{\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathcal{R}^m}{\textbf{argmax}}\ 1^T\boldsymbol{\alpha} - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

$$\text{s.t. } \sum_{j=1}^{m}\alpha_j y_j = 0,$$

$$\alpha_j = C - \beta_j,\ \ \alpha_j, \beta_j \geq 0$$

$$\text{for } j = 1, ..., m$$

Inference: given an input $\boldsymbol{x}$

$$f(x) = \boldsymbol{f}^T\hat{\boldsymbol{w}} + \hat{b}$$

Inference: given an input $\boldsymbol{x}$

$$f(x) = \underbrace{\sum_{j=1}^{m}\hat{\alpha}_j y'_j \mathcal{K}(\boldsymbol{x}'_j, \boldsymbol{x})}_{\boldsymbol{w}^T\boldsymbol{f}} + \hat{b}$$

Note.

- The dual form $K(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is the only place that involves with $h(\cdot)$.
- $h(\cdot)$ can be replaced with a generalized form of kernel functions $K(\boldsymbol{x}_i, \boldsymbol{x}_j)$.
- But $K(\boldsymbol{x}_i, \boldsymbol{x}_j)$ has to satisfy a certain condition (next).

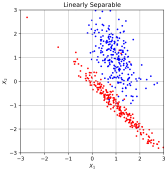
Primal
Inference: given an input $\boldsymbol{x}$
$$f(\boldsymbol{x}) = \boldsymbol{w}^T h(\boldsymbol{x}) + \hat{b}$$

Dual (Kernel)
Inference: given an input $\boldsymbol{x}$
$$f(\boldsymbol{x}) = \sum_{j=1}^{m} \hat{\alpha}_j y'_j \mathcal{K}(x'_j, \boldsymbol{x}) + \hat{b}$$

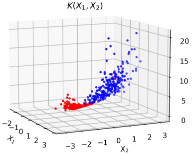
Condition: a kernel function $K(\boldsymbol{x}_i, \boldsymbol{x}_j) : \mathcal{R}^{d \times d} \Rightarrow \mathcal{R}^{1 \times 1}$ is symmetric and positive semi-definite:

$$K(\boldsymbol{x}_j, \boldsymbol{x}) = K(\boldsymbol{x}, \boldsymbol{x}_j) \quad \text{and} \quad K(\boldsymbol{x}_j, \boldsymbol{x}_j) \geq 0$$

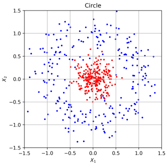
- **Linear.** $K(\boldsymbol{x}_j, \boldsymbol{x}) = \boldsymbol{x}_j^T \boldsymbol{x}$ is the similarity of a pair using Pearson (standard) correlation.
- **Polynomial.** $K(\boldsymbol{x}_j, \boldsymbol{x}) = (\gamma \cdot \boldsymbol{x}_j^T \boldsymbol{x} + \beta)^p$ where p is a positive integer and β is a coefficient.
- **Radial basis function (RBF).** $K(\boldsymbol{x}_j, \boldsymbol{x}) = e^{-\gamma ||\boldsymbol{x} - \boldsymbol{x}_j||_2^2}$ where $\gamma > 0$.
- **Hyperbolic tangent.** $K(\boldsymbol{x}_j, \boldsymbol{x}) = \tanh(\gamma \cdot \boldsymbol{x}_j^T \boldsymbol{x} + \beta)$.



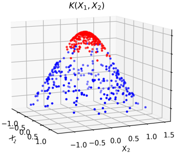
Polynomial $\Rightarrow \mathcal{K}(\boldsymbol{x}_1, \boldsymbol{x}_2) = (\gamma \cdot \boldsymbol{x}_1^T \boldsymbol{x}_2 + \beta)^d \Rightarrow$

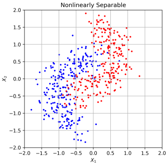


$\sqrt{(\underline{x}_1 - \underline{x}_2)^2}^2 = \|x_1\|_2^2 - 2x_1^T x_2 + \|x_2\|_2^2$

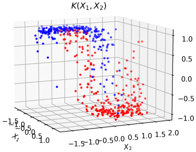


Radial Basis $\Rightarrow \mathcal{K}(\boldsymbol{x}_1, \boldsymbol{x}_2) = e^{-\gamma \cdot ||\boldsymbol{x}_1 - \boldsymbol{x}_2||_2^2} \Rightarrow$

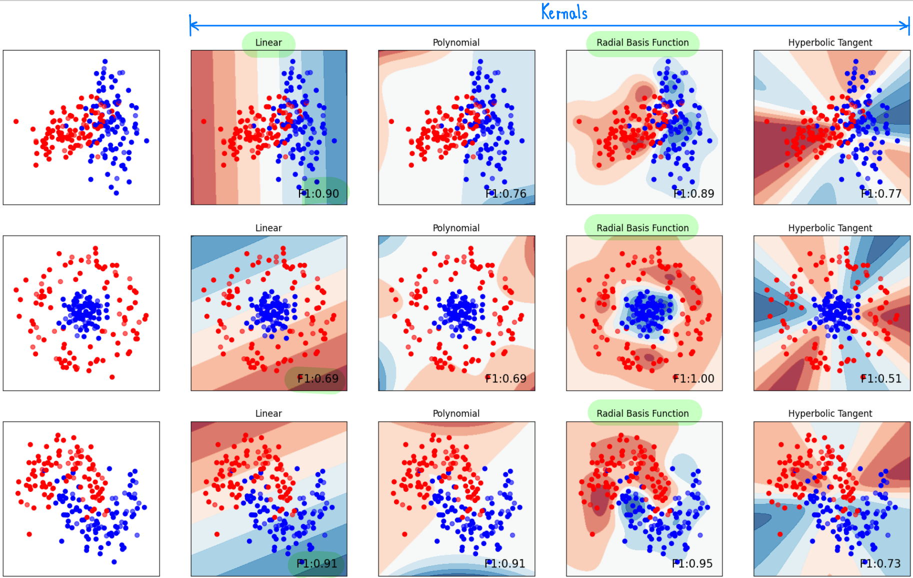




Tanh $\Rightarrow \mathcal{K}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \tanh(\gamma \cdot \boldsymbol{x}_1^T \boldsymbol{x}_2 + \beta) \Rightarrow$



Kernel functions and classification



1 . มักจะนึกถึงก่อน

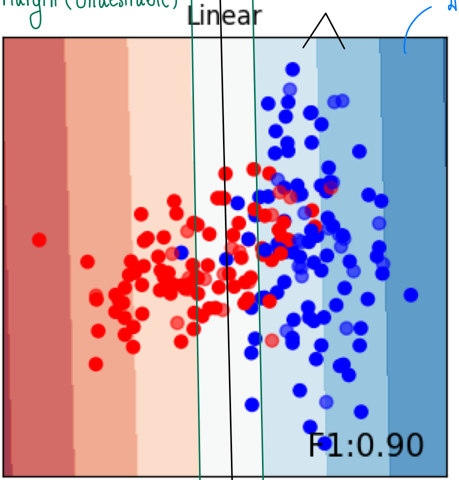
2 . อันดับรอง.

Kernel functions and classification

Margin (Undesirable)

ไม่ค่อยมั่นใจมากนัก.

มั่นใจแน่ๆ ว่าเป็น
Class สีฟ้า



Hyperplane

How to choose SVM parameters?

$f(\underline{x}) = \text{Kernal } f^n + C \cdot \sum_{j=1}^{m} z_j$

C is the penalty parameter common to all choices of kernel:

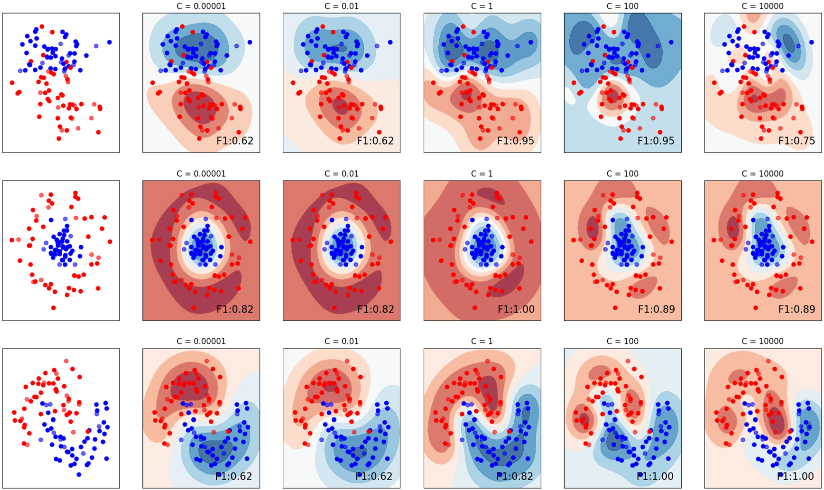
- **Low C:** SVM will allow the inclusion of the wrong sides and can be very rough (high bias), if too small. $\Rightarrow$ Underfitting to training data.
- **High C:** SVM will strictly classify the data according to its label and can exhibit high variance, if too large. $\Rightarrow$ Overfitting to training data.

γ is the decay rate of RBF in $e^{-\gamma||\boldsymbol{x}-\boldsymbol{z}||_2^2}$

- γ can be seen as **the inverse of radius of the influence area that**:
  a training point has the same influence $\Rightarrow \gamma_{\text{small}}||\boldsymbol{x}_i - \boldsymbol{x}_j||_2^2 = \gamma_{\text{large}}||\boldsymbol{x}_i - \boldsymbol{x}_k||_2^2$.
- **Low γ:** influence area covers a far single training point can reach and affect the model.
  $\Rightarrow$ Underfitting to training data.
- **High γ:** influence area covers only a close single training point can reach.
  $\Rightarrow$ Overfitting to training data.

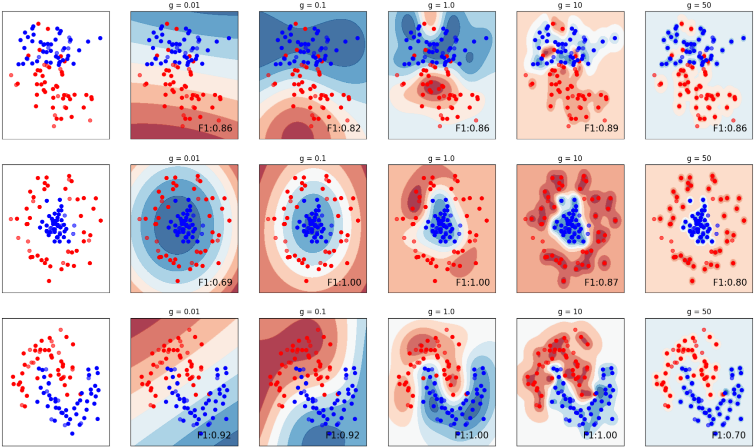
These two parameters affect SVM's performance (typically chosen via cross-validation).

Effect of RBF parameters (varying C)



- If C is too small, SVM will allow the inclusion of the wrong sides $\Rightarrow$ underfitting.
- If C is too large, SVM will strictly classify the data according to its label $\Rightarrow$ overfitting.

Effect of RBF parameters (varying γ)



- If γ is too small, the model cannot capture the complexity of data $\Rightarrow$ underfitting.
- If γ is too large, the model considers only the support vector itself $\Rightarrow$ overfitting.
- Intermediate γ gives smooth models that detect data pattern; can be made more complex by increasing C.

[1] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY, USA: Springer New York Inc., 2001.

[2] Gareth James et al. *An Introduction to Statistical Learning: with Applications in R*. Springer, 2013. URL: https://faculty.marshall.usc.edu/gareth-james/ISL/.

[3] Jitkomut Songsiri. "Statistical Learning". In: in EE575 Random Processes for EE. Chulalongkorn University, 2022. Chap. 9. Support vector machine. URL: http://jitkomut.eng.chula.ac.th/ee575.html.